ENS3553/ENS6160 Signals a

ENS3553/ENS6160 Signals and Systems

Laboratory Experiment 2

Frequency Response of Systems – Lab Results

Student Name:_____ Student No:_____

Post-lab

A. Frequency response of an RC circuit

a) Determining the frequency response,

b) Calculating | |

f	(rad/s)	| |	| | in dB
1 kHz	$= 2\pi(1000)$ $= 6283.19$ rad/s	0.9956	-0.0384
10 kHz	62831.85	0.7277	-2.76
40 kHz	251327.41	0.2564	-73.71

c) Plot of amplitude and phase response

nd Systems

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d) Questions

a) What kind of filtering is performed by this RC circuit?

b) What does -3db equate to in terms of voltage amplitude ratio?

c) What is the cut-off frequency (-3dB point on the plot)?

d) How can we determine this value from the circuit parameters?

e) Describe the behaviour of this system in the frequency domain and in the time domain.

B. Response of the RC circuit to an AC voltage using the frequency response

a) Theoretical output voltage

f	(rad/s)	$\lvert \quad \rvert$	$\angle$	$\lvert \quad \rvert$	$\angle$
1 kHz	6283.19	0.9956	-5.38	7.467	-5.38
10 kHz	62831.85	0.7277	-43.30	5.458	-43.30
40 kHz	251327.41	0.2564	-75.14	1.923	-75.14

b) Magnitude and phase spectra of output voltage

C. Response of the RC circuit to square wave using the frequency response

a) Fourier Series coefficients

vp=

$$V_p = \frac{15}{2} = 7.5\,V$$

$$f_0 = 8000\,Hz$$

Hz

Fourier Coefficient (	Value	Frequency
b_1	9.5493 V	8,000 Hz
b_3	3.1831 V	24,000 Hz
b_5	1.9099 V	40,000 Hz

(b) and (c) Magnitude of frequency response and Magnitude of harmonics

f (Hz)	w	|hw |	Input magnitude (		Output magnitude	
			V	dBV	V	dBV
Fundamental =	$50265.48\,rad/s$	0.7985	9.5493	19.60	7.625	17.65
3rd harmonic =	150,796.45	0.4041	3.183	10.06	1.286	2.18
5th harmonic =	251,327.41	0.2564	1.910	5.62	0.490	−6.20

d) Description of output waveform

Results Sheet

A. Verify frequency response in time domain

1. Table A: Theoretical frequency response vs measured frequency response

Frequency	Magnitude		Phase	
	Theoretical Value	**Measured Value**	**Theoretical Value**	**Measured Value**
1 kHz	0.9956	0.9744	−5.38°	−6.65°
10 kHz	0.7277	0.7233	−43.30°	−42.30°
40 kHz	0.2564	0.2705	−75.14°	$-73.71°$

2. Sketch of input-output pairs.

3. Filtering type and comparison of cut-off frequency (-3dB point) between the measured and theoretical values.

B. Verify frequency response in frequency domain

1. Table B: Frequency response (magnitude) for sinusoids

Frequency	Output Magnitude	Input Magnitude	Magnitude Gain	Theoretical Response
1 kHz	17.62	17.84	−0.23 dB	−0.04dB
10 kHz	15.19	18.01	−2.81 dB	−2.76dB
40 kHz	6.49	17.84	−11.36	−11.82 dB

2. Comparison of measured results with theoretical (post-lab) values

3. Sketches of output spectra and description of comparison

C. Response to square wave input

1. Table C: Square wave theoretical vs measured harmonic response

Spectrum characteristic	Output Magnitude	Input Magnitude	Magnitude Response	Theoretical Response
Fundamental	dBV	dBV	dB	dB

3rd harmonic	dBV	dBV	dB	dB
5th harmonic	dBV	dBV	dB	dB

2. Sketch of output spectra

3. Sketch of input and output time domain waveforms and explanation

Discussion of experimental results

~ End of Lab Report ~